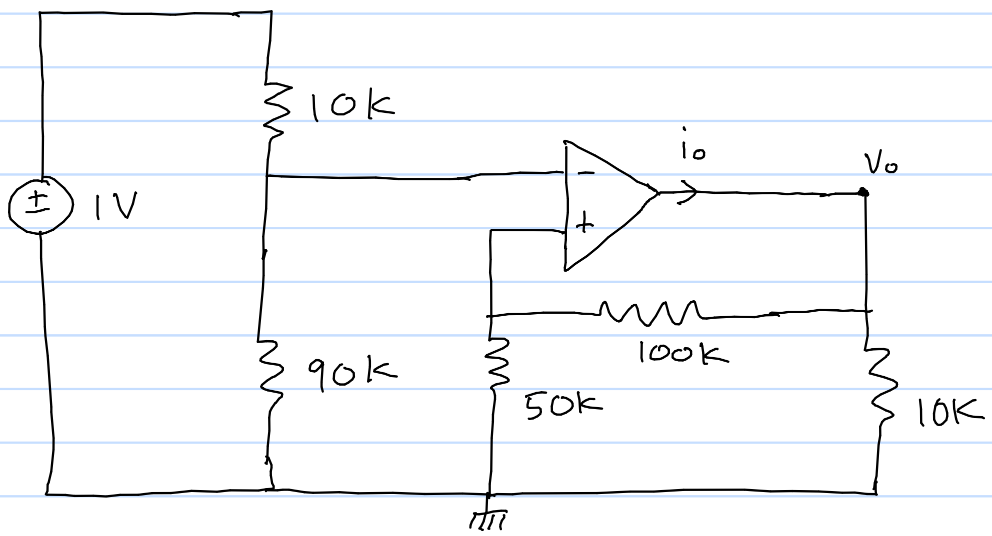
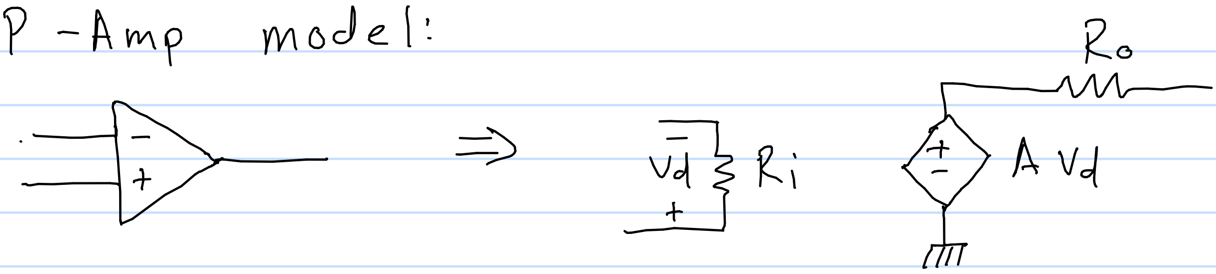
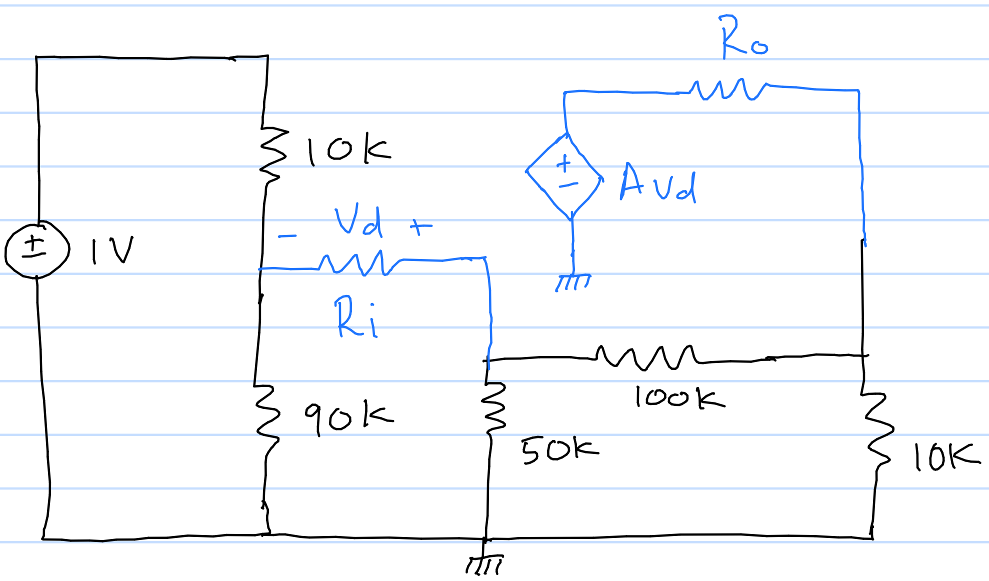
5.13 Find V_o and i_o in the circuit

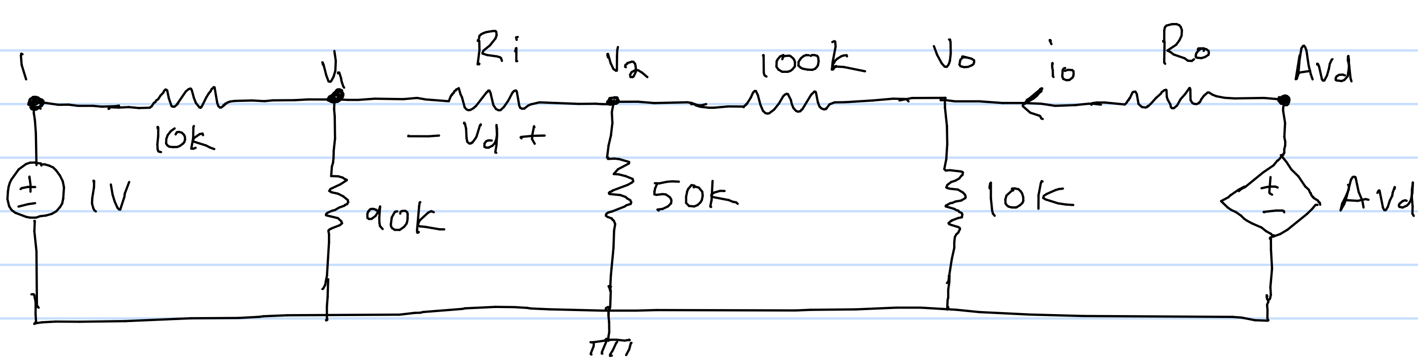


OP-Amp model:



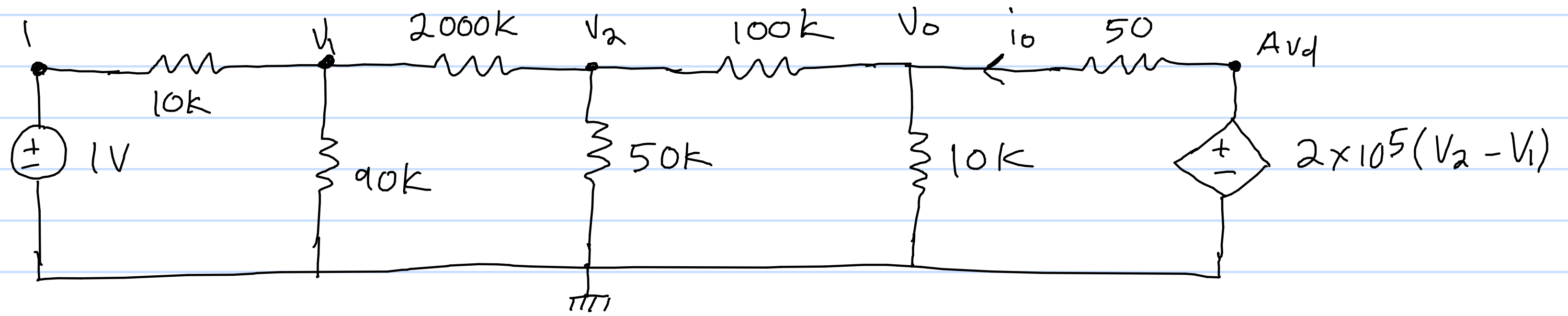
Redraw Circuit:





using parameters from 741 op-amp:

$A = 2 \times 10^5$ $R_i = 2 \times 10^6\ \Omega$ $R_o = 50$



Nodal analaysis:

@ V_1:
$$\frac{V_1-1}{10\times10^3}+\frac{V_1}{90\times10^3}+\frac{V_1-V_2}{2\times10^6}=0$$

$$200V_1+\frac{200V_1}{9}+V_1-V_2=200$$

$$\frac{2009V_1}{9}-V_2=200$$

@ V_2:
$$\frac{V_1-V_2}{2\times10^6}+\frac{V_2}{50\times10^3}+\frac{V_2-V_o}{100\times10^3}=0$$

$$V_1-V_2+40V_2+20V_2-20V_o=0$$

$$V_1+59V_2-20V_o=0$$

@ V_o:
$$\frac{V_o-V_2}{100\times10^3}+\frac{V_o}{10\times10^3}+\frac{V_o-Avd}{50}=0$$

$$\frac{V_o-V_2}{100\times10^3}+\frac{V_o}{10\times10^3}+\frac{V_o-2\times10^5(V_2-V_1)}{50}=0$$

$$V_o-V_2+10V_o+2000V_o-400\times10^6(V_2-V_1)=0$$

$$400\times10^6V_1-400000001V_2+2011V_o=0$$

$$\frac{2009V_1}{9} - V_2 = 200$$

$$V_1 + 59V_2 - 20V_0 = 0$$

$$400\times10^6 V_1 - 400000001 V_2 + 2011 V_0 = 0$$

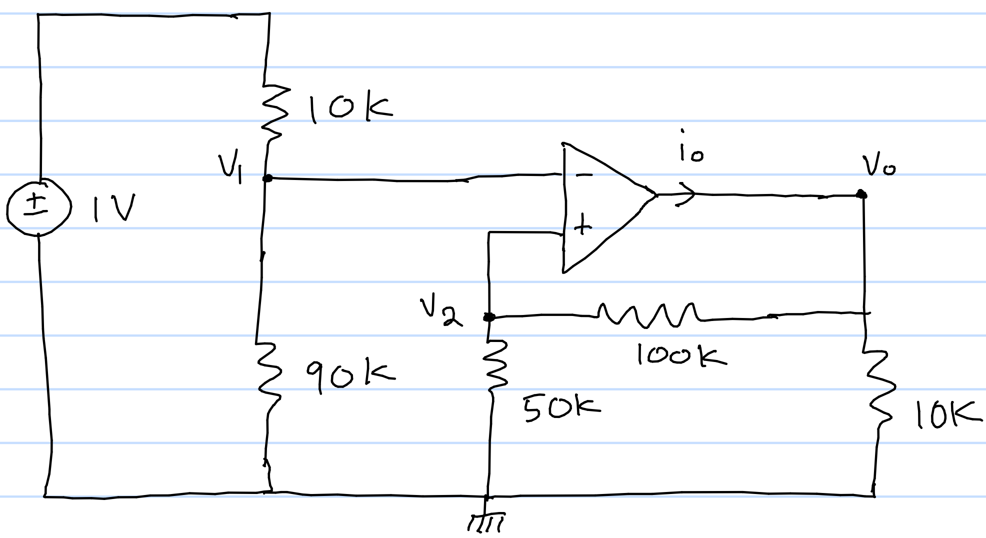
Using a calculating to solve equations:

$V_1 = 0.900000611$

$V_2 = 0.9000136333$

$V_0 = 2.700040221$

5.13 Find V_o and i_o in circuit:



ideal op-amp $\quad V_1 = V_2 \quad i_1 = i_2 = 0$

V_1 using voltage division:

$$V_1 = \frac{1 \times 90 \times 10^3}{10 \times 10^3 + 90 \times 10^3} = 0.9\ V$$

V_2 using voltage division:

$$V_2 = \frac{V_o \times 50 \times 10^3}{50 \times 10^3 + 100 \times 10^3}$$

$$V_2 = \frac{V_o}{3} \qquad \text{since } V_2 = V_1$$

$$0.9 = \frac{V_o}{3}$$

$$V_o = 2.7$$

OP-Amp model vs ideal model:

$$2.700040221 - 2.7 = 0.000040221$$
$$\approx 0$$